Question: `{question}`
Please reason step by step, and put your final answer within `boxed{}`.

J Color thickets

Beyond language models, we observe similar effects in diffusion models. Here, one can think of "color thickets," where certain regions of parameter space preferentially generate images with specific color palettes or visual styles. These differences may not reflect fundamentally distinct generative mechanisms, yet they still produce dense clusters of high-scoring samples under color- or style-sensitive metrics.

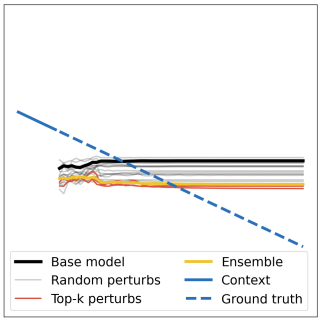
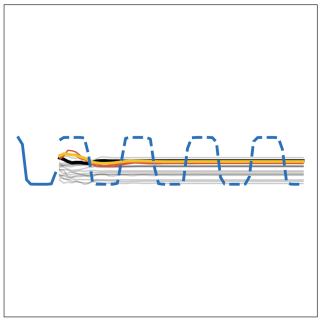
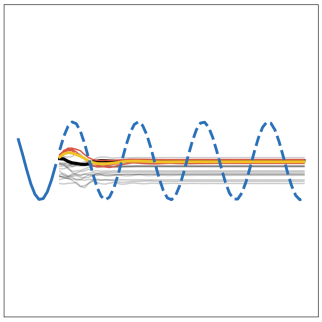
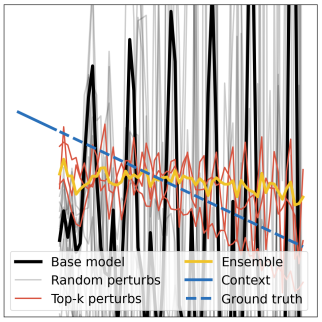
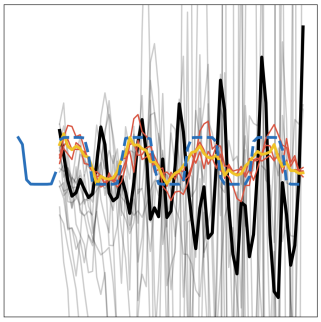
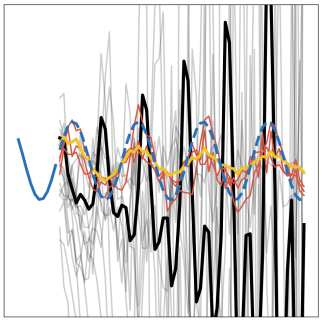
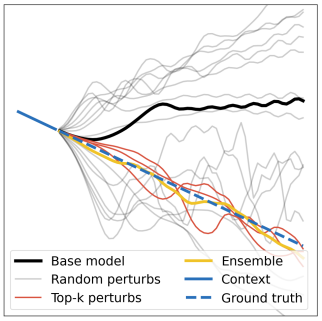
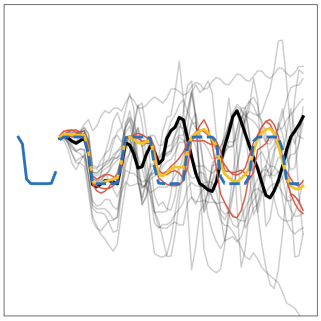
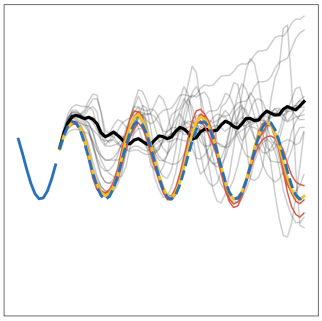
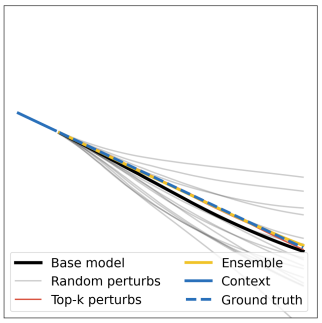
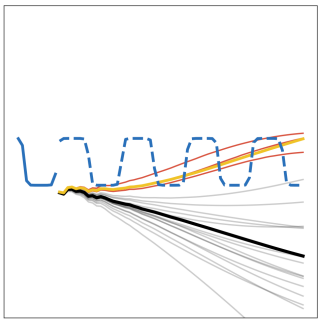
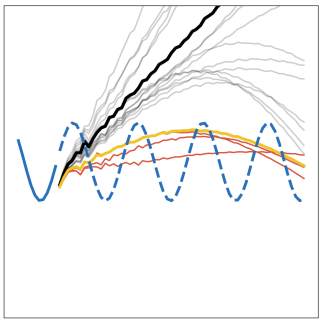
For instance, if the evaluation rewards blue-dominant images, a region that consistently generates bluish outputs forms a "blue thicket." More generally, thickets may arise not only from differences in reasoning, but also from biases in generative tendencies such as color, texture, or style.

Table 4: Experimental results on reasoning benchmarks. We compare RandOpt against RL-based methods (PPO, GRPO), Evolution Strategies (ES), Best-of-N and test-time majority voting (TT-MV) across model scales. Results are averaged over 3 runs with standard deviation shown in gray. **Bold** indicates best performance, underlined indicates runner-up. Details of the hyperparameters for all methods are provided in Appendix E.3.

Model	Method	Countdown (math) ↑	GSM8k (math) ↑	MATH-500 (math) ↑	OlyBench (math) ↑	MBPP (prog.) ↑	ROCStories (writing) ↑	USPTO (chemistry) ↑
Qwen2.5-0.5B-Inst	Base	$0.1_{\pm 0.1}$	$39.9_{\pm 0.0}$	$19.8_{\pm 0.0}$	$4.2_{\pm 0.4}$	$30.9_{\pm 0.2}$	$22.0_{\pm 0.0}$	$29.9_{\pm 0.3}$
	TT-MV†	$0.3_{\pm 0.3}$	$41.0_{\pm 0.1}$	$29.2_{\pm 0.5}$	$6.7_{\pm 0.2}$	$37.0_{\pm 0.3}$	$19.5_{\pm 0.2}$	$30.2_{\pm 0.2}$
	Best-of-N‡	$5.9_{\pm 2.0}$	$40.5_{\pm 1.4}$	$34.5_{\pm 0.5}$	$10.5_{\pm 0.3}$	$34.9_{\pm 1.3}$	$20.3_{\pm 0.3}$	$26.8_{\pm 1.2}$
	PPO	$14.8_{\pm 2.8}$	$43.2_{\pm 1.2}$	$33.4_{\pm 1.7}$	$16.1_{\pm 0.0}$	$37.8_{\pm 3.0}$	$19.1_{\pm 0.8}$	$31.2_{\pm 0.6}$
	GRPO	$13.0_{\pm 0.0}$	$48.4_{\pm 0.8}$	$33.7_{\pm 0.9}$	$6.9_{\pm 0.1}$	$42.8_{\pm 3.7}$	$30.9_{\pm 1.1}$	$31.7_{\pm 0.2}$
	ES	$\mathbf{14.9}_{\pm 1.6}$	$42.6_{\pm 0.7}$	$30.5_{\pm 0.7}$	$\mathbf{16.4}_{\pm 0.2}$	$45.1_{\pm 3.8}$	$32.0_{\pm 0.8}$	$31.6_{\pm 0.3}$
	RandOpt	$8.4_{\pm 0.3}$	$54.1_{\pm 0.8}$	$35.3_{\pm 0.6}$	$15.8_{\pm 0.7}$	$46.2_{\pm 0.4}$	$\mathbf{32.2}_{\pm 0.4}$	$32.2_{\pm 0.9}$
	ES + TT-MV	$11.2_{\pm 0.8}$	$\mathbf{61.2}_{\pm 0.5}$	$\mathbf{41.3}_{\pm 0.5}$	$13.5_{\pm 0.7}$	$\mathbf{48.9}_{\pm 0.7}$	$30.5_{\pm 0.9}$	$\mathbf{32.5}_{\pm 0.4}$
Qwen2.5-1.5B-Inst	Base	$6.7_{\pm 0.1}$	$58.8_{\pm 0.2}$	$43.2_{\pm 0.1}$	$13.4_{\pm 0.4}$	$62.3_{\pm 0.2}$	$46.7_{\pm 0.1}$	$30.2_{\pm 0.5}$
	TT-MV†	$30.8_{\pm 0.4}$	$69.1_{\pm 0.3}$	$50.0_{\pm 0.4}$	$15.9_{\pm 0.5}$	$68.0_{\pm 0.3}$	$43.5_{\pm 0.5}$	$33.0_{\pm 0.4}$
	Best-of-N‡	$19.5_{\pm 0.2}$	$65.4_{\pm 1.9}$	$53.0_{\pm 0.6}$	$20.0_{\pm 0.3}$	$69.5_{\pm 0.3}$	$42.2_{\pm 0.8}$	$32.7_{\pm 1.1}$
	PPO	$27.0_{\pm 0.0}$	$71.6_{\pm 0.7}$	$55.9_{\pm 0.3}$	$26.3_{\pm 0.1}$	$68.5_{\pm 0.4}$	$51.8_{\pm 0.8}$	$31.9_{\pm 0.2}$
	GRPO	$27.5_{\pm 0.7}$	$72.1_{\pm 0.7}$	$54.1_{\pm 0.5}$	$18.8_{\pm 0.8}$	$\mathbf{70.2}_{\pm 0.4}$	$53.6_{\pm 1.3}$	$31.8_{\pm 0.0}$
	ES	$44.2_{\pm 0.0}$	$71.7_{\pm 0.9}$	$54.1_{\pm 2.8}$	$27.2_{\pm 1.2}$	$69.9_{\pm 0.6}$	$\mathbf{60.2}_{\pm 0.6}$	$32.0_{\pm 0.2}$
	RandOpt	$\mathbf{52.7}_{\pm 0.5}$	$76.4_{\pm 0.3}$	$59.7_{\pm 0.6}$	$\mathbf{30.4}_{\pm 0.7}$	$69.6_{\pm 0.5}$	$48.5_{\pm 0.7}$	$\mathbf{34.3}_{\pm 0.5}$
	ES + TT-MV	$39.7_{\pm 0.3}$	$\mathbf{80.4}_{\pm 0.8}$	$\mathbf{60.7}_{\pm 0.7}$	$28.9_{\pm 0.7}$	$\mathbf{70.2}_{\pm 0.6}$	$59.1_{\pm 0.2}$	$32.2_{\pm 0.6}$
Qwen2.5-3B-Inst	Base	$10.0_{\pm 0.1}$	$79.8_{\pm 0.4}$	$58.6_{\pm 0.2}$	$24.5_{\pm 0.2}$	$69.5_{\pm 0.3}$	$54.7_{\pm 0.1}$	$38.5_{\pm 0.5}$
	TT-MV†	$12.8_{\pm 0.4}$	$82.5_{\pm 0.2}$	$60.8_{\pm 0.2}$	$21.8_{\pm 0.3}$	$74.5_{\pm 0.5}$	$57.3_{\pm 0.4}$	$43.2_{\pm 0.7}$
	Best-of-N‡	$28.5_{\pm 1.2}$	$83.3_{\pm 1.4}$	$62.5_{\pm 0.8}$	$28.0_{\pm 0.6}$	$73.0_{\pm 0.9}$	$55.0_{\pm 0.7}$	$44.3_{\pm 1.0}$
	PPO	$35.3_{\pm 0.4}$	$83.1_{\pm 0.2}$	$64.1_{\pm 1.1}$	$34.4_{\pm 0.2}$	$76.3_{\pm 1.0}$	$49.0_{\pm 0.6}$	$44.7_{\pm 5.8}$
	GRPO	$32.6_{\pm 0.1}$	$83.2_{\pm 0.2}$	$64.6_{\pm 1.0}$	$29.0_{\pm 0.0}$	$77.0_{\pm 0.9}$	$56.3_{\pm 4.4}$	$49.7_{\pm 2.1}$
	ES	$55.6_{\pm 0.5}$	$85.8_{\pm 5.1}$	$61.9_{\pm 0.3}$	$36.4_{\pm 0.1}$	$\mathbf{77.2}_{\pm 1.2}$	$\mathbf{64.5}_{\pm 0.6}$	$\mathbf{52.9}_{\pm 1.0}$
	RandOpt	$58.4_{\pm 0.2}$	$87.1_{\pm 0.8}$	$\mathbf{68.7}_{\pm 0.7}$	$39.2_{\pm 0.6}$	$75.9_{\pm 0.6}$	$56.5_{\pm 0.3}$	$42.3_{\pm 0.3}$
	ES + TT-MV	$\mathbf{61.9}_{\pm 0.8}$	$\mathbf{87.9}_{\pm 0.9}$	$67.7_{\pm 0.7}$	$\mathbf{39.7}_{\pm 0.4}$	$76.3_{\pm 1.1}$	$55.0_{\pm 0.4}$	$39.8_{\pm 0.3}$
OLMo3-7B-Inst	Base	$64.8_{\pm 0.2}$	$82.9_{\pm 0.4}$	$60.6_{\pm 0.1}$	$28.7_{\pm 0.1}$	$65.9_{\pm 0.2}$	$64.0_{\pm 0.3}$	$27.2_{\pm 0.4}$
	TT-MV†	$66.8_{\pm 0.3}$	$81.4_{\pm 0.4}$	$61.5_{\pm 0.5}$	$26.5_{\pm 0.4}$	$60.5_{\pm 0.5}$	$63.2_{\pm 0.4}$	$30.2_{\pm 0.5}$
	Best-of-N‡	$67.5_{\pm 0.8}$	$85.0_{\pm 1.2}$	$63.0_{\pm 0.7}$	$30.5_{\pm 0.5}$	$64.0_{\pm 0.9}$	$63.5_{\pm 0.6}$	$32.0_{\pm 1.0}$
	PPO	$69.0_{\pm 0.1}$	$88.4_{\pm 0.4}$	$63.1_{\pm 0.3}$	$28.0_{\pm 0.2}$	$67.7_{\pm 0.2}$	$64.7_{\pm 1.6}$	$40.2_{\pm 3.2}$
	GRPO	$68.5_{\pm 0.7}$	$87.0_{\pm 0.2}$	$63.5_{\pm 0.4}$	$27.9_{\pm 0.6}$	$70.8_{\pm 2.2}$	$65.8_{\pm 0.6}$	$\mathbf{51.0}_{\pm 0.3}$
	ES	$71.0_{\pm 0.8}$	$87.2_{\pm 0.2}$	$69.9_{\pm 1.0}$	$33.1_{\pm 0.8}$	$72.0_{\pm 0.5}$	$65.7_{\pm 1.4}$	$45.2_{\pm 1.0}$
	RandOpt	$\mathbf{85.0}_{\pm 0.7}$	$89.5_{\pm 0.2}$	$\mathbf{73.7}_{\pm 0.4}$	$35.4_{\pm 0.4}$	$\mathbf{75.1}_{\pm 0.9}$	$64.5_{\pm 0.3}$	$43.0_{\pm 0.5}$
	ES + TT-MV	$75.6_{\pm 0.7}$	$\mathbf{90.2}_{\pm 0.6}$	$62.0_{\pm 1.1}$	$\mathbf{44.7}_{\pm 0.3}$	$72.5_{\pm 0.6}$	$\mathbf{66.0}_{\pm 0.8}$	$46.3_{\pm 0.6}$
OLMo3-7B	Base	$9.8_{\pm 0.4}$	$78.5_{\pm 0.4}$	$31.3_{\pm 0.3}$	$13.9_{\pm 0.3}$	$29.1_{\pm 0.3}$	$24.1_{\pm 0.1}$	$29.0_{\pm 0.3}$
	TT-MV†	$11.5_{\pm 0.3}$	$79.5_{\pm 0.2}$	$36.2_{\pm 0.2}$	$15.7_{\pm 0.2}$	$39.1_{\pm 0.3}$	$42.2_{\pm 0.2}$	$35.4_{\pm 0.2}$
	Best-of-N‡	$18.0_{\pm 0.5}$	$82.0_{\pm 1.0}$	$45.0_{\pm 0.8}$	$20.5_{\pm 0.4}$	$38.0_{\pm 1.2}$	$50.0_{\pm 0.8}$	$38.0_{\pm 0.9}$
	PPO	$21.9_{\pm 0.7}$	$82.8_{\pm 0.9}$	$51.8_{\pm 0.7}$	$22.9_{\pm 0.5}$	$57.9_{\pm 2.0}$	$64.8_{\pm 0.3}$	$\mathbf{49.8}_{\pm 0.3}$
	GRPO	$28.8_{\pm 0.1}$	$78.2_{\pm 0.3}$	$52.0_{\pm 0.9}$	$6.9_{\pm 0.6}$	$58.5_{\pm 2.8}$	$62.2_{\pm 2.8}$	$48.0_{\pm 0.5}$
	ES	$26.0_{\pm 0.3}$	$\mathbf{89.1}_{\pm 0.6}$	$61.0_{\pm 4.9}$	$30.2_{\pm 0.4}$	$\mathbf{61.8}_{\pm 2.3}$	$64.4_{\pm 1.0}$	$48.1_{\pm 1.5}$
	RandOpt	$\mathbf{30.2}_{\pm 0.2}$	$85.0_{\pm 0.3}$	$59.3_{\pm 0.5}$	$28.9_{\pm 0.5}$	$40.5_{\pm 0.2}$	$64.5_{\pm 0.3}$	$44.3_{\pm 0.3}$
	ES + TT-MV	$23.6_{\pm 0.2}$	$86.4_{\pm 0.5}$	$\mathbf{69.0}_{\pm 0.5}$	$\mathbf{43.3}_{\pm 0.2}$	$56.8_{\pm 0.6}$	$\mathbf{76.3}_{\pm 0.6}$	$38.1_{\pm 0.8}$
Llama3.1-8B-Inst	Base	$10.8_{\pm 0.2}$	$79.8_{\pm 0.3}$	$47.0_{\pm 0.0}$	$19.2_{\pm 0.3}$	$56.4_{\pm 0.0}$	$51.8_{\pm 0.0}$	$19.5_{\pm 0.3}$
	TT-MV†	$25.6_{\pm 0.3}$	$72.2_{\pm 0.4}$	$49.2_{\pm 0.3}$	$24.5_{\pm 0.5}$	$59.5_{\pm 0.4}$	$53.2_{\pm 0.4}$	$25.4_{\pm 0.2}$
	Best-of-N‡	$40.0_{\pm 1.0}$	$83.5_{\pm 1.2}$	$52.0_{\pm 0.9}$	$23.0_{\pm 0.6}$	$60.0_{\pm 1.0}$	$54.0_{\pm 0.8}$	$28.0_{\pm 1.1}$
	PPO	$9.9_{\pm 0.1}$	$81.6_{\pm 4.7}$	$45.5_{\pm 2.0}$	$16.1_{\pm 0.6}$	$55.2_{\pm 1.5}$	$57.8_{\pm 1.3}$	$32.9_{\pm 4.6}$
	GRPO	$10.0_{\pm 0.2}$	$80.2_{\pm 1.1}$	$45.1_{\pm 1.4}$	$23.7_{\pm 0.9}$	$61.0_{\pm 1.9}$	$62.2_{\pm 2.8}$	$35.0_{\pm 0.7}$
	ES	$60.3_{\pm 0.5}$	$82.4_{\pm 0.7}$	$39.1_{\pm 1.7}$	$22.3_{\pm 0.5}$	$64.8_{\pm 0.8}$	$\mathbf{68.1}_{\pm 5.5}$	$34.1_{\pm 2.0}$
	RandOpt	$\mathbf{63.6}_{\pm 0.4}$	$86.7_{\pm 0.6}$	$59.5_{\pm 0.8}$	$\mathbf{32.1}_{\pm 0.0}$	$\mathbf{65.2}_{\pm 0.9}$	$59.0_{\pm 0.7}$	$41.0_{\pm 0.6}$
	ES + TT-MV	$62.5_{\pm 0.9}$	$\mathbf{87.5}_{\pm 0.6}$	$\mathbf{62.0}_{\pm 0.7}$	$\mathbf{32.1}_{\pm 0.5}$	$63.1_{\pm 0.8}$	$65.2_{\pm 0.7}$	$\mathbf{64.5}_{\pm 0.7}$

†TT-MV: Test-Time Majority Vote over test samples from a single trained model with different seeds.
‡Best-of-N: Pass@k metric with k=50.

Table 5: RandOpt on 1D signals (approximation). Each cell corresponds to a pretraining–post-training pair. In this experiment we plot results on the same function that we fit to during post-training; this is a test of apprpoximation only, not generalization to new functions. We also compare to Xavier initialization (Glorot & Bengio, 2010) and Kaiming initialization He et al. (2015). For each pretraining method, we pick a perturbation noise scale, σ that is large enough to show functional variation; this value is shown underneath the pretraining method name.

Pretraining \ Post-training	One linear function	One square wave	One sinusoid
None (Xavier init) $\sigma = 0.05$			
None (Kaiming init) $\sigma = 0.05$			
Mixed $\sigma = 0.002$			
Linear $\sigma = 0.002$